CSD 342 Data Mining and Data Warehousing

Assignment

Total Marks: 20

Questions 1-5 are 2 marks each.

Question 6 is for 10 marks.

1. Download data set Iris and calculate the following:
    a. the average (mean) value for each of the four features
    b. the standard deviation for each of the features
    c. repeat steps (a) and (b) but separately for each type of flower
    d. draw four box plots, one for each feature, such that each figure shows three boxes, one for each type of flower. Properly label your axes in all box plots.
    e. Find out outliers (if any) in the dataset

    Data set Iris can be downloaded from http://archive.ics.uci.edu/ml/datasets/iris.

2. Suppose that the data mining task is to cluster the following eight points (with (x, y) representing location) into three clusters:
    A1(2, 10), A2(2, 5), A3(8, 4), B1(5, 8), B2(7, 5), B3(6, 4), C1(1, 2), C2(4, 9):
    The distance function is Euclidean distance. Suppose initially we assign A1, B1, and C1 as the center of each cluster, respectively. Use separately k-means, k-mediods and k-median algorithms to show the following:
    (a) The three cluster centers after the first round execution
    (b) The final three clusters

3. For the k-means algorithm, it is interesting to note that by choosing the initial cluster centers carefully, we may be able to not only speed up the algorithm's convergence, but also guarantee the quality of the final clustering. The k-means++ algorithm is a variant of k-means, which chooses the initial centers as follows. First, it selects one center uniformly at random from the objects in the data set. Iteratively, for each object p other than the chosen center, it chooses an object as the new center. This object is chosen at random with probability proportional to dist.(p)^2, where dist(p) is the distance from p to the closest center that has already been chosen. The iteration continues until k centers are selected. Explain why this method will not only speed up the convergence of the k-means algorithm, but also guarantee the quality of the final clustering results.

4. Given two objects represented by the tuples (22, 1, 42, 10) and (20, 0, 36, 8):
    (a) Compute the Euclidean distance between the two objects.
    (b) Compute the Manhattan distance between the two objects.
    (c) Compute the Minkowski distance between the two objects, using q = 3.

5. Explain the DBSCAN algorithm. Show 3 examples where DBSCAN is a better choice compared to k-means.

6. Refer to the paper attached along with ths assignment and answer the following questions.
    a. What are the challenges in performing clustering over large dynamic networks?
    b. Explain a real life application of the problem discussed in the paper.
    c. Explain the static P2P k-means algorithm and show how the algorithm can be modified in a dynamic environment.
    d. How are the results analyzed and what can you conclude from the paper.
    e. Based on the future work suggest modifications in the algorithm that may improve performance.